Literature Review: AI-Powered Personalized Learning and Concept Mastery Platform

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***Abstract*—Large Language Models (LLMs) are now used in many education tools, but most of them answer one question at a time and do not remember what the student knows. This review studies five works related to our capstone project: two surveys on LLMs in education, one scoping review on practical and ethical challenges, one survey on Knowledge Tracing (KT), and one system that joins KT with Retrieval-Augmented Generation (RAG). From these works we find three main gaps: hallucination, no memory of student skill, and weak ethical design. We then explain how our platform tries to reduce these gaps.**

***Index Terms*—Large language models, personalized learning, knowledge tracing, retrieval-augmented generation, education**

I. Introduction

Many universities and companies now add AI to their learning systems so that students can understand hard concepts and perform better. LLMs such as ChatGPT, Gemini and Claude can explain topics, make questions and give feedback at any time. A student, however, may still not know which concepts they really understand and what to study next. Our project is an AI-powered learning platform that uses the student's own study material, tracks concept mastery, and recommends the next step.

This review covers five papers. Section II discusses LLMs in education. Section III discusses practical and ethical challenges. Section IV explains knowledge tracing. Section V discusses TutorLLM, the closest work to ours. Section VI gives the research gap and our solution.

II. LLMs in Education

Xu et al. [1] give a broad survey of LLMs for education (LLMEdu). They show that LLMs help with personalized learning support, feedback on assessments, real-time tutoring, content generation and teacher assistance. They also mention guided learning, where the model asks questions instead of giving the final answer. Khanmigo, built by OpenAI and Khan Academy, is given as an example of this style.

The same survey also lists problems. LLMs can produce false information, which is a serious risk when students cannot judge the answer. Students may depend too much on the tool and think less by themselves. Privacy of student data and unequal access to AI tools are also named as open issues. The authors suggest RAG with a verified knowledge source as one way to reduce wrong answers.

III. Practical and Ethical Challenges

Yan et al. [2] reviewed 118 peer-reviewed papers and found 53 use cases of LLMs in education, grouped in nine categories such as grading, feedback, prediction, content generation and recommendation. Their main finding is that most work is still early research. About 89 studies were at the applied research level, and none had reached the level of a verified prototype in real classrooms.

They also found problems in how the research is reported. Most studies (107) did not share enough detail to be replicated. Most systems (109) were transparent only to AI researchers and not to teachers or students. Privacy was rarely discussed, and most tools worked only for English. The authors recommend a human-centred approach, open sharing of models and data, and use of stronger models. This supports our decision to show evidence for answers and to explain every recommendation to the student.

IV. Knowledge Tracing

Knowledge Tracing (KT) estimates how a student's knowledge changes over time, using the history of answered exercises [3]. Shen et al. divide KT models into three groups. Bayesian models, such as Bayesian Knowledge Tracing [6], treat knowledge as a hidden state that is either learned or not learned. Logistic models, such as Performance Factor Analysis, predict the chance of a correct answer from past successes and failures. Deep learning models, such as Deep Knowledge Tracing [7], use neural networks to model complex learning, and they usually predict better.

The survey also points out that deep KT models are hard to interpret. It is difficult to know how the hidden state represents real mastery, so students and teachers may not trust the result. The survey further explains that KT needs a lot of data, and that data is often sparse in real systems. Finally, it suggests that LLMs could help KT by reading the student's open-ended answers.

This matches our design. Our Milestone-1 plan starts with a simple and explainable mastery score for each concept, and a deep KT model is kept only as a later extension when enough data is available.

TABLE I
GAPS IN EXISTING WORK AND PROPOSED SOLUTION

Finding	Limitation	Our Solution
LLMs give false answers [1], [4]	Large documents and single-query answers can lose context	RAG on uploaded material with source evidence and a quiz-verification loop
LLMs have no native memory of skill [3], [4]	Past quizzes and weak topics are not remembered	Memory with LangGraph stored in a database; concept mastery score
Ethical and transparency problems [2]	Unrestricted chatbots can reveal answers and are hard to explain	Two-layer guardrail, guided (Socratic) questioning, reasons shown for each recommendation

V. RETRIEVAL-AUGMENTED GENERATION AND TUTORLLM

RAG joins a language model with an external source of text, so the answer can be based on retrieved content and not only on the model memory [5]. In education, the uploaded course material can be this source.

Li et al. [4] proposed TutorLLM, which combines KT and RAG with GPT-4. It has three parts: a KT model (MLFBK) that tracks the learning state, a Scraper that collects text from online course pages, and a RAG-based LLM that gives answers and study recommendations. They tested it with 30 first-year undergraduate students for two weeks in a linear algebra course. The TutorLLM group had the highest mean score (74.48), compared with 72.81 for the hybrid group and 71.97 for the general LLM group. However, the ANOVA test ($p = 0.462$) showed that the difference was not statistically significant. The usability score was good (SUS average 76.35), and students spent 36% more time with TutorLLM.

TutorLLM shows that joining KT with an LLM is possible. Its limits are the small sample, the short study, and the dependence on course web pages instead of the student's own files. It also does not clearly handle concept extraction or answer safety.

VI. RESEARCH GAP AND PROPOSED SOLUTION

Table I compares the problems found in the reviewed work with our proposed solution.

One more issue is that checking a topic as done does not prove mastery. If a system marks a concept as mastered after one correct answer, it can hide a real weakness. For this reason, our platform keeps a mastery score for each concept using several signals, and a concept with no attempts is marked unknown instead of weak. We plan to test the agent with DeepEval and to measure retrieval and quiz quality.

VII. CONCLUSION

The reviewed work shows that LLMs are useful for learning but have problems with wrong answers, missing memory of the learner, and weak ethical design. KT gives a way to model the learner, and RAG gives a way to ground answers. TutorLLM shows the two can work together, but its results are not yet strong. Our platform aims to join document-grounded RAG, concept-level mastery tracking and explainable recommendations in one learning loop. Its real value will be shown in the evaluation phase.

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